Using GaN Transistors In Power Converters

The basic requirements for power semiconductors are efficiency, reliability, controllability, and cost-effectiveness. Breakthroughs in wide-bandgap semiconductors, especially in gallium nitride (GaN), have enabled devices with high conductivity and hyper-fast switching, with a silicon-like cost structure and fundamental operating mechanism. This article discusses some design aspects for implementing GaN transistors in power converters

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With the emerging technologies like electric vehicles, there is a need for power converters which are highly efficient and have high power density. Silicon devices based power converters are limited in terms of efficiency and power density, as silicon MOSFETs cannot switch faster than around 500kHz frequency. Thus arises the need for wide bandgap semiconductors that feature high voltage operation, high current handling capacity, are low on resistance, have lower switching loss, and high frequency operation.

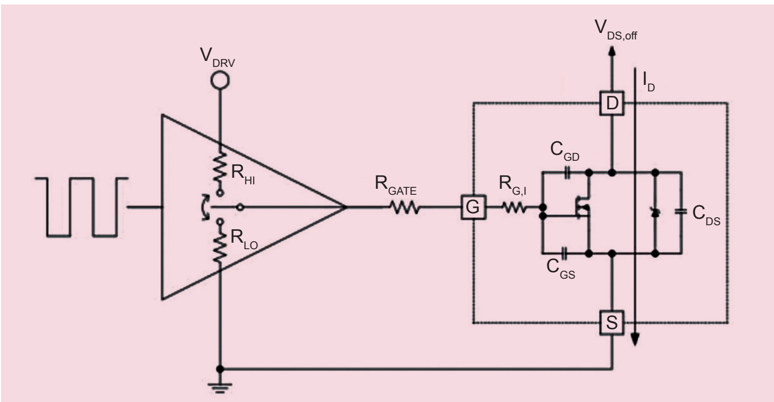

Fig. 1: Common gate elements in the gate drive path (Source: Texas Instruments application note 'External Gate Resistor Design Guide for Gate Drivers')

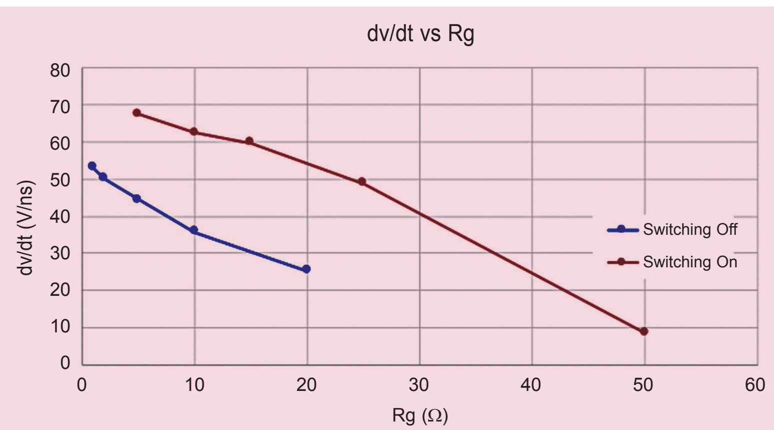

Fig. 2: dv/dt vs gate resistance

Silicon carbide and gallium nitride are two most popular wide bandgap materials. Gallium nitride or GaN transistors are currently widely used for applications ranging from electric vehicles to phone chargers. High power density of GaN devices allows compact power circuits and fast power delivery.

What makes these GaN devices special is their low 'on' resistance and very low gate capacitance. The low on resistance and low gate-drive capacitance makes it possible for GaN devices to switch hundreds of volts in nanoseconds, giving it multiple megahertz capability. This capability leads to smaller power converters and higher fidelity class D amplifiers.

The switching frequency is also limited by the series gate resistance of the transistor. The gate resistance limits how quickly the capacitance of a field effect transistor can be charged or discharged. The metal gates enable GaN to have gate resistances of a couple of tenths of an ohm. This low gate resistance also helps with dV/dt immunity.

However, two major problems are common in the application of wide-bandgap devices: voltage overshoots and oscillations.